

Day 5 - Testing, Error Handling, aur Backend Integration Refinement

Marketplace (E-Commerce)

Functional Testing – Products Page

All products load successfully from the backend API. Product details (images, titles, prices) display correctly. Filters and search functionality are under development and will be implemented in the next iteration.

Figure 1: Products Page (all items loaded)

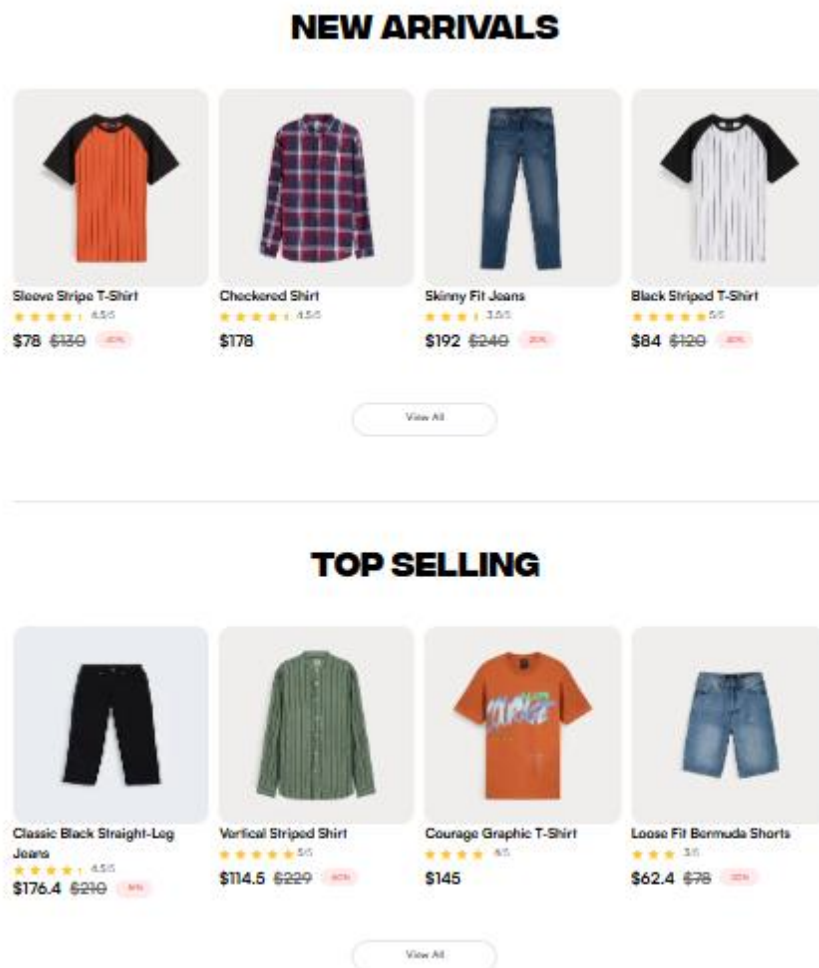


Figure 2: Products Listing Page (all items loaded)

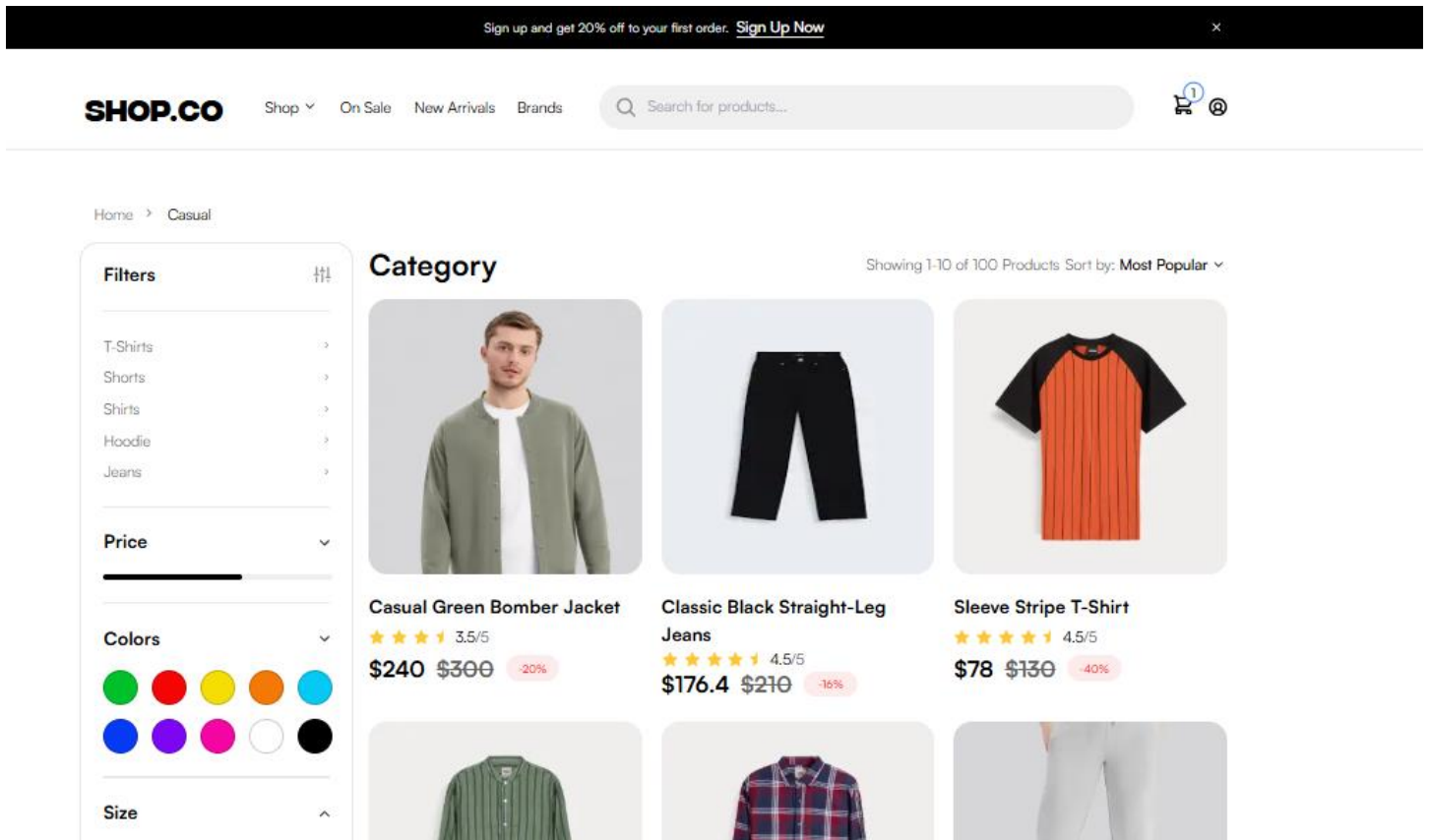
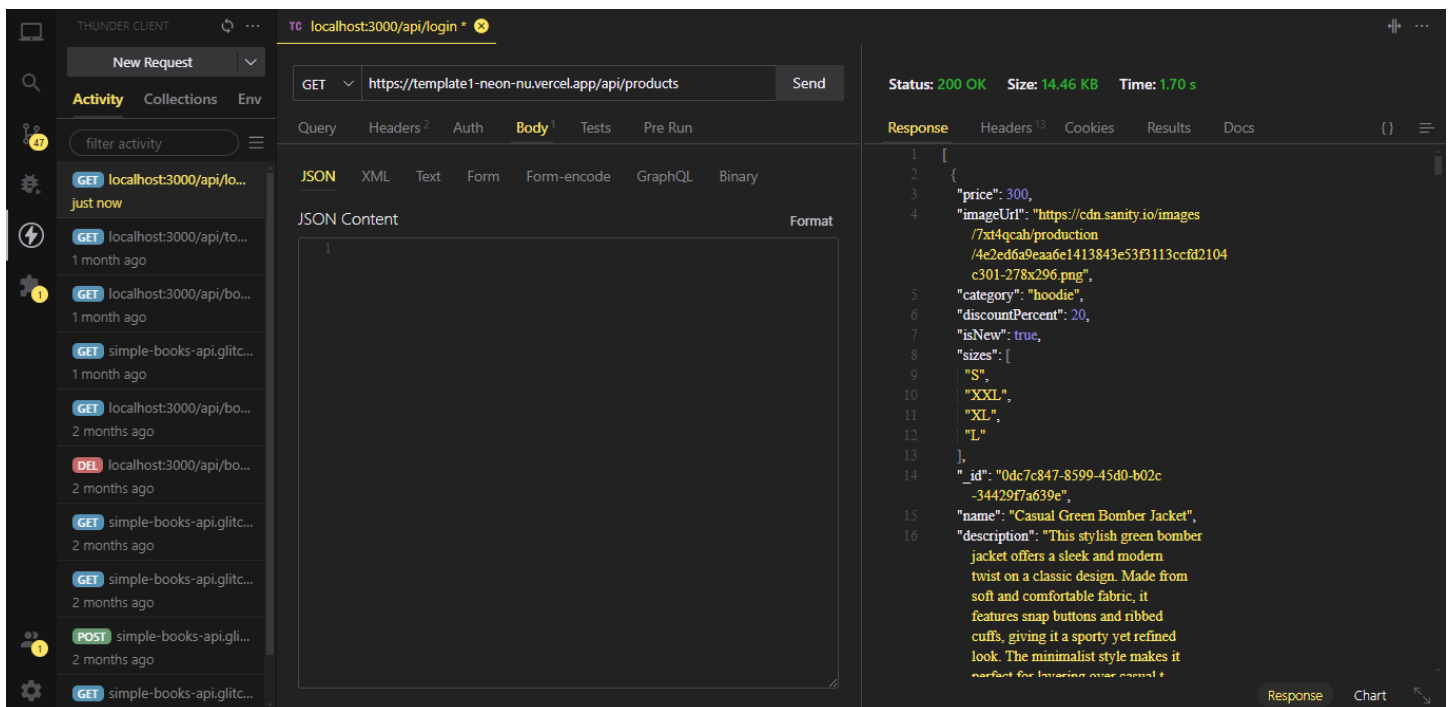


Figure 3: Thunder Client Api Response (200 OK)



Functional Testing - Cart Operations

Cart functionality works as expected. Items can be added, removed, and the total price updates dynamically. The checkout page is fully functional with a smooth navigation flow."

Figure: 4 Product Detail Page (Product adding in cart)

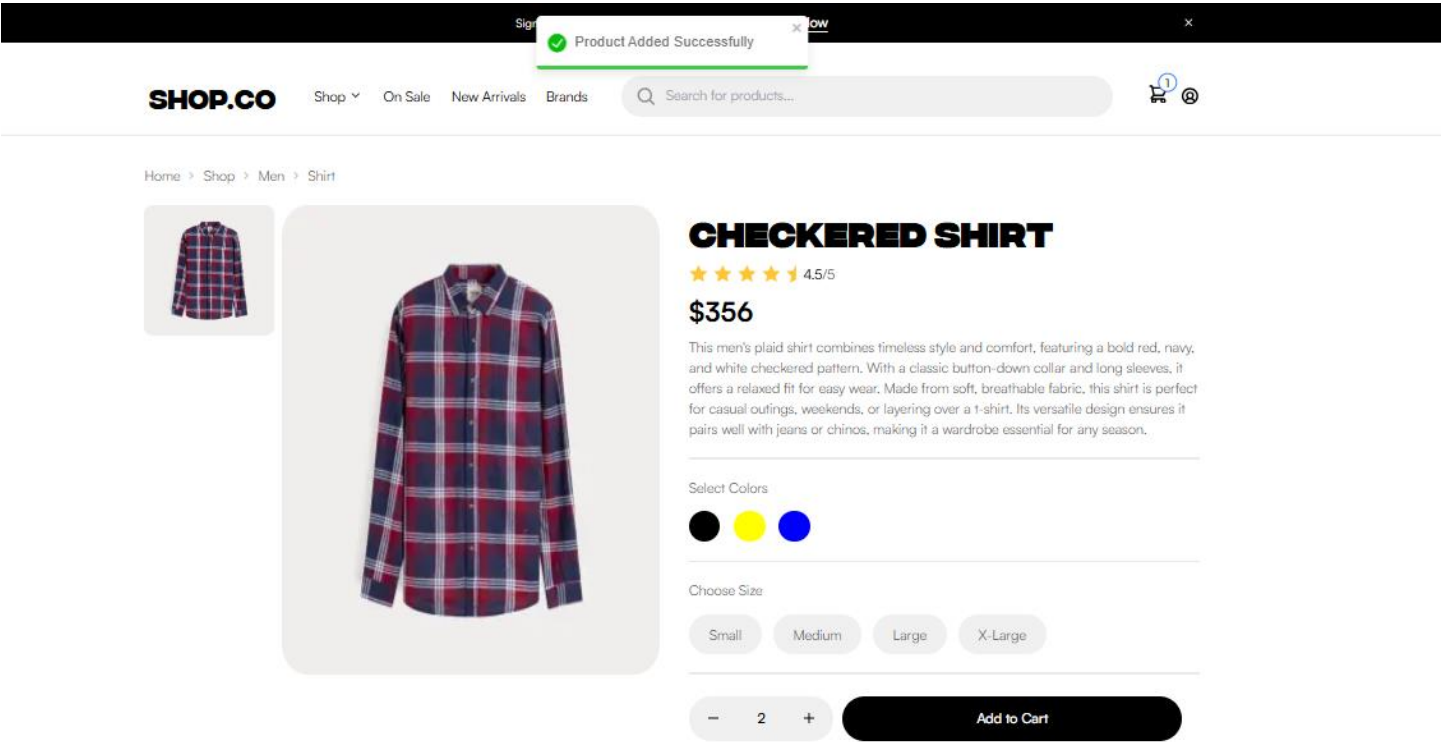


Figure: 5 Cart Page with Items.

YOUR CART

Home > Cart



Checked Shirt
Size: Medium
Color: Blue

\$356

-

2

+

Order Summary

Subtotal

\$356

Discount (-20%)

-\$0

Delivery Fee

TBD

Total

\$356

Add promo code

Apply

Go to Checkout →

Figure 6: Checkout Page

SHOP.CO

Shop ▾On SaleNew ArrivalsBrands

Q

Search for products...

1

Shopping Cart

Home > Cart > Checkout

Order Summery



Checked Shirt
Color: Blue Size: Medium
QTY: 2

\$356

Total:

\$356

Billing Information

First Name

Enter your first name

Last Name

Enter your last name

Email

Enter your email

City

Enter your city

Address

Enter your address

Phone

Enter your phone number

Zip Code

Enter your zip code

Place Order

[Testing Report Sample.csv](#)

Test Case ID	Test Case Description	Test Steps	Expected Result	Actual Result	Status	Severity Level	Assigned To	Remarks
TC001	Validate product listing page	/products	Products displayed correctly	Products displayed correctly	Passed	High	-	No issues found
TC002	Test cart functionality	1. Add product to cart 2. Visit cart page	Cart updates with added product	Cart updates as expected	Passed	High	-	Works as expected
TC003	Check checkout process	/checkout	Order placed successfully	Order placed successfully	Passed	High	-	Payment gateway not implemented
TC004	Test API error handling	Disconnect API > Refresh page	Show fallback UI with error message	Error message shown in console log	Failed	low	-	Working unexpectedly failed
TC005	Ensure responsiveness on mobile	Resize browser window	Layout adjusts properly to screen size	Responsive layout not working	Failed	Low	-	Working Failed
TC006	Test login/signup pages	1. Go to /login 2. Go to /signup	Login/Signup pages displayed	Pages not implemented	Failed	High	-	Pending development
TC007	Test filters on product listing	1. Apply filters (e.g., price range) 2. Check filtered results	Filtered products displayed	Filters not implemented	Failed	Medium	-	Pending development

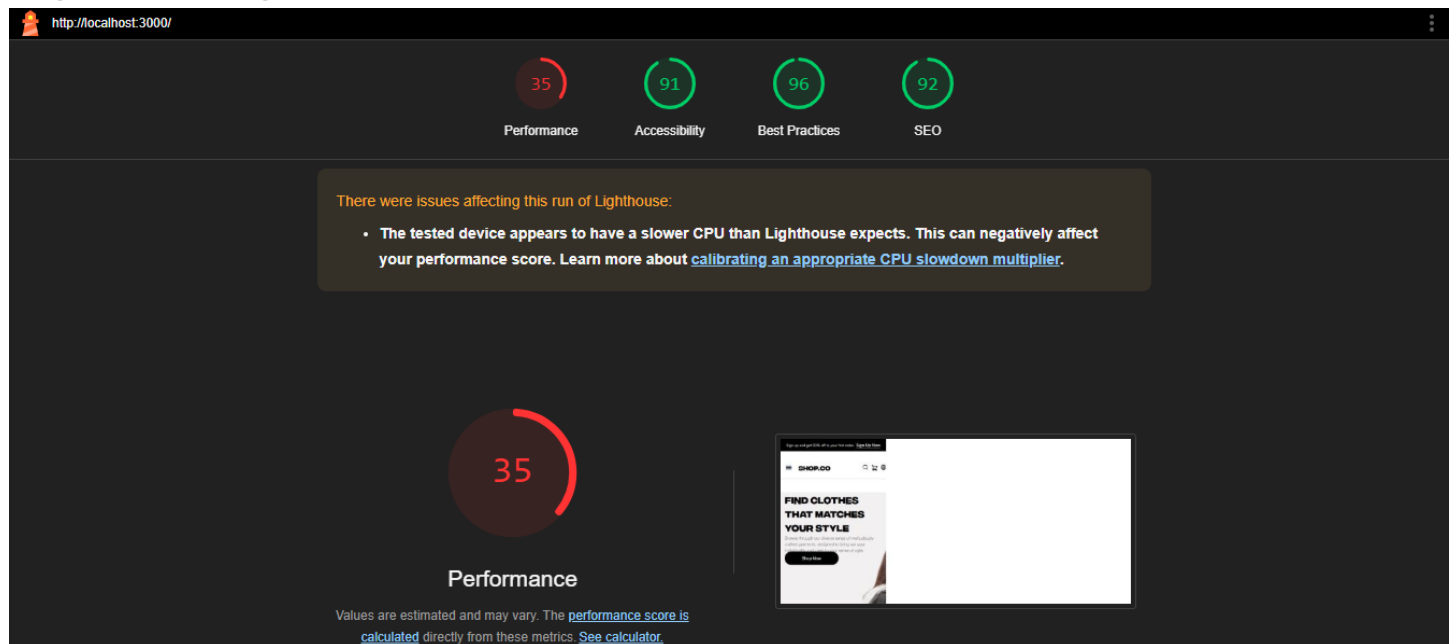
Performance Testing

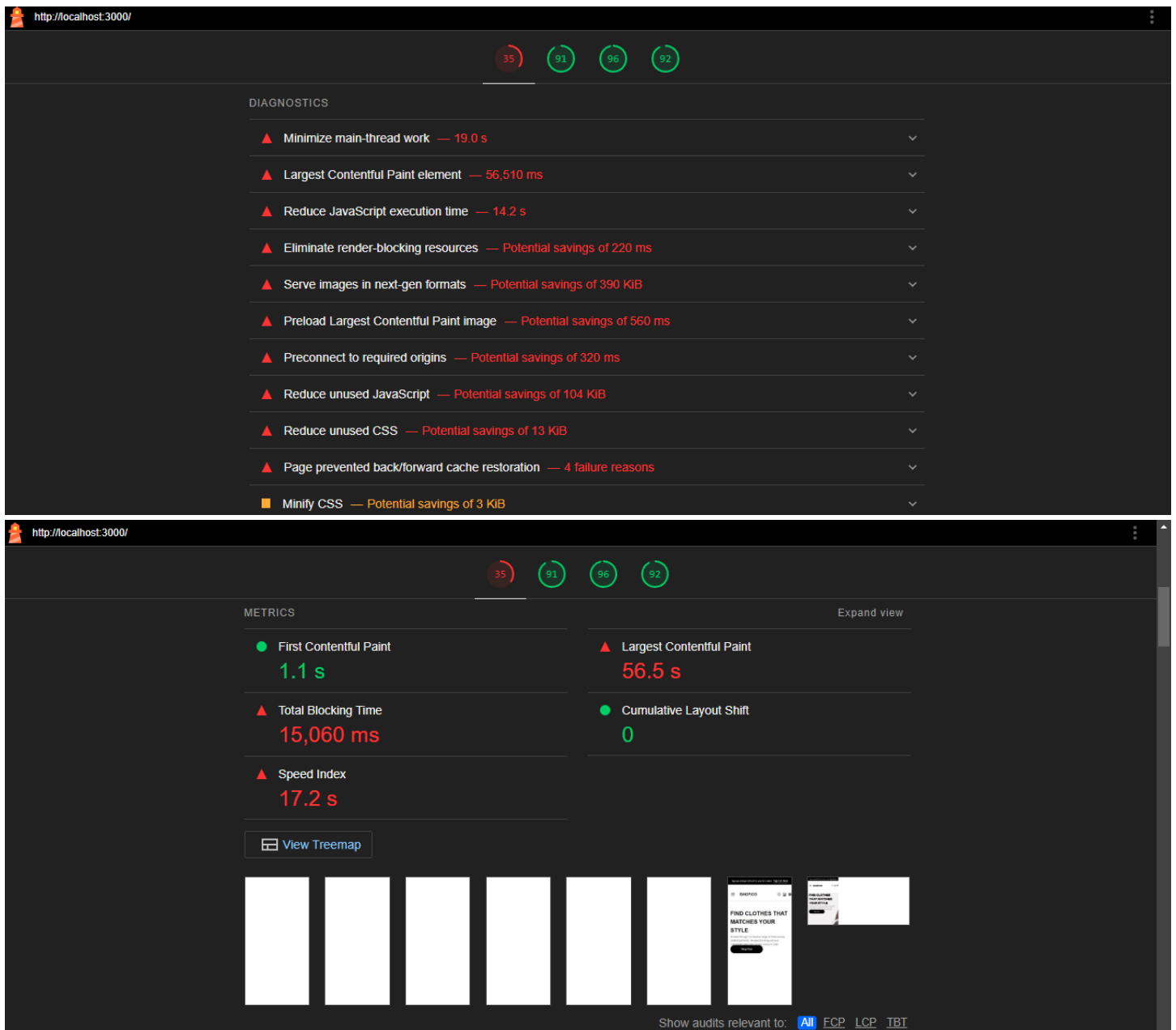
Lighthouse is an open-source tool that analyzes a website's **performance, accessibility, SEO, and best practices**. It is integrated into **Google Chrome DevTools** and runs **automated audits** to help improve site speed, user experience, and search engine ranking. Lighthouse evaluates **Core Web Vitals**, including page load speed, interactivity, and visual stability. It also

detects issues related to **Progressive Web Apps (PWA)**, security, and overall site health. The tool generates a detailed report with **scores and recommendations**, allowing developers to optimize their websites effectively

Lighthouse Report - Performance

Figure 1: Lighthouse Performance Score (35/100)





Lighthouse Report – Accessibility

Figure 2: Lighthouse Accessibility Score (91/100)

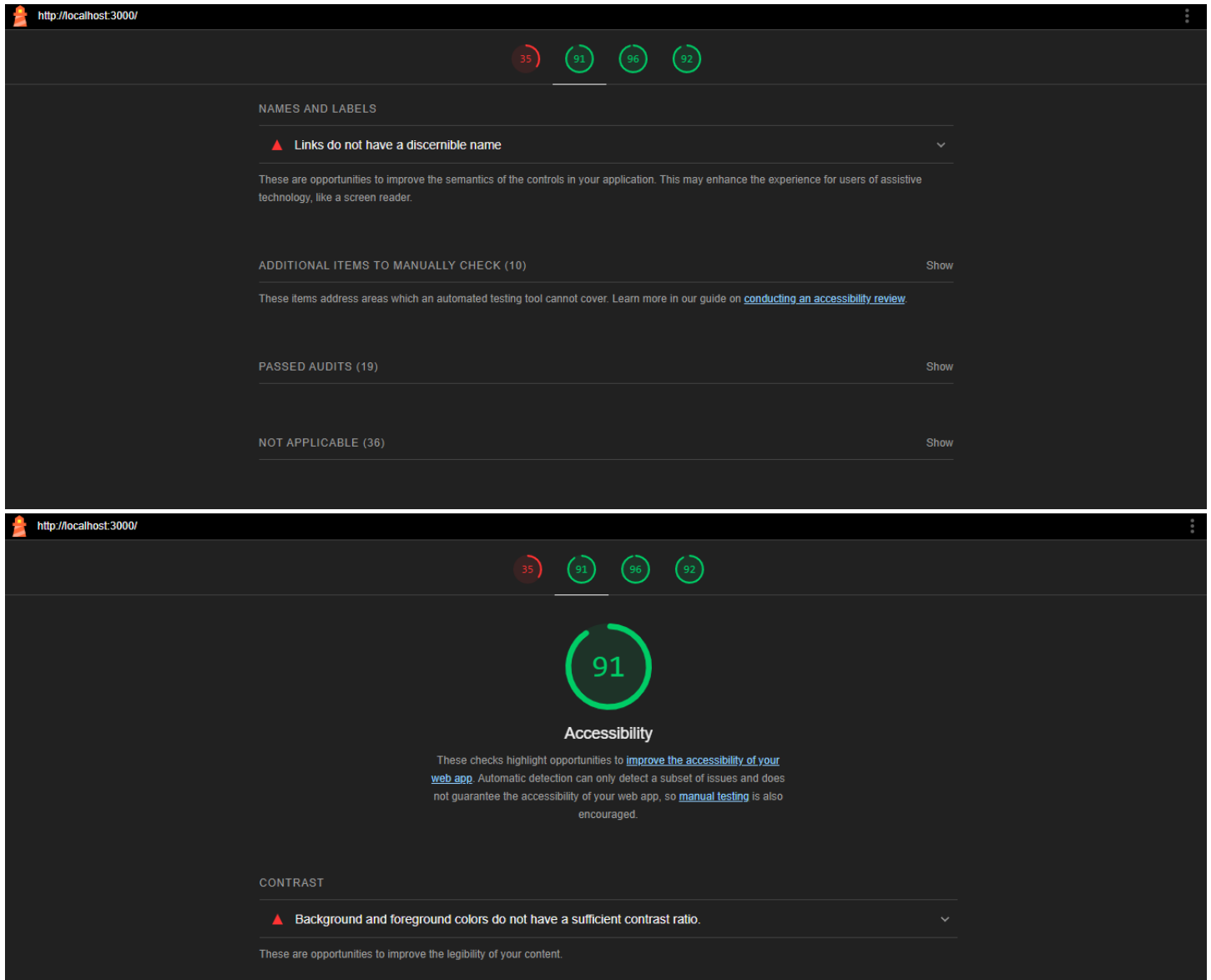
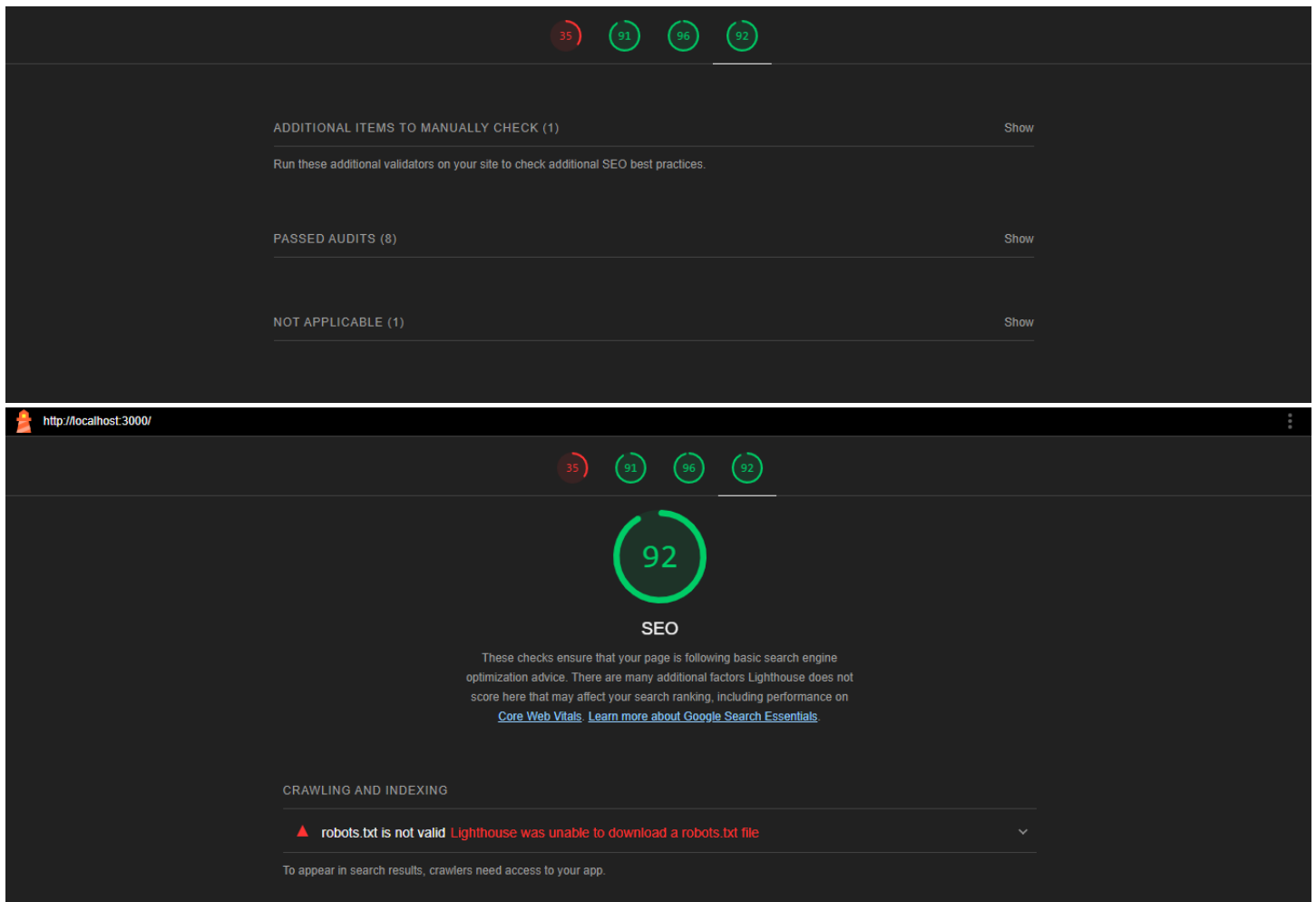
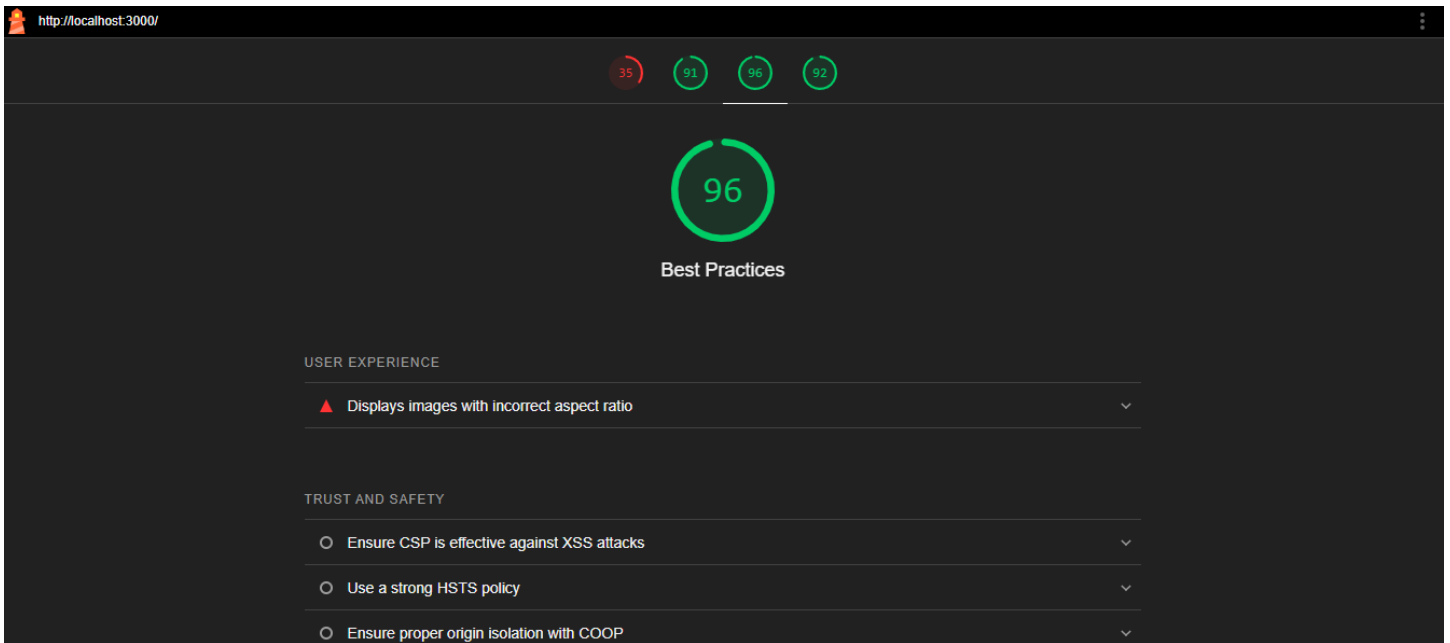


Figure 3: Lighthouse SEO Score (92/100)



[Lighthouse Report – Best Practices](#)

Figure 4: Lighthouse Best Practices Score (96/100)



Conclusion

In this document, I conducted **functional testing** to evaluate the core features of the website, including **cart functionality, checkout process, and product display**. These features were working correctly as expected. However, I have not yet implemented **product listing with filtered functionality, login/signup, or payment integration**, but I plan to complete them in the future.

During the testing process, I faced some challenges:

- **Lighthouse Usage:** While Lighthouse provided valuable insights, interpreting some of the metrics—especially related to performance and accessibility—required additional understanding.
- **Testing Complex Scenarios:** Some functionalities required real-user interactions, which Lighthouse alone could not fully simulate.
- **Identifying Bugs:** Pinpointing the exact cause of certain minor issues required manual debugging alongside automated testing.

Despite these challenges, this testing helped me understand the current state of the website and provided a roadmap for future improvements.

